Tree trunks & generated bark

Continuous trunks and a complete set of five image-generated bark materials.



Reported in the old shortcut build



Actual forest in the updated renderer · cropped

Different scene and camera. The close material comparisons on the next page use matched views.

Trunk seams: retains the preceding continuous-ring trunk fix, smooth taper normals and matching wrap tangents. This revision replaces the remaining scanned bark without changing the tree geometry.

Bark: all five bark types now use image-generated sources: smooth, fissured, scaly, palm-fibrous and peeling. Linear-light exposure retains colour detail; repeat edges and approximate relief normals are prepared offline.

Delivery: the shortcut selects the verified candidate at handoff. An already running old session needs to be relaunched to show the changes.

Matched material views

The same specimen, camera and production shader: previous scanned materials versus generated replacements.

PREVIOUS SCANNED BARK

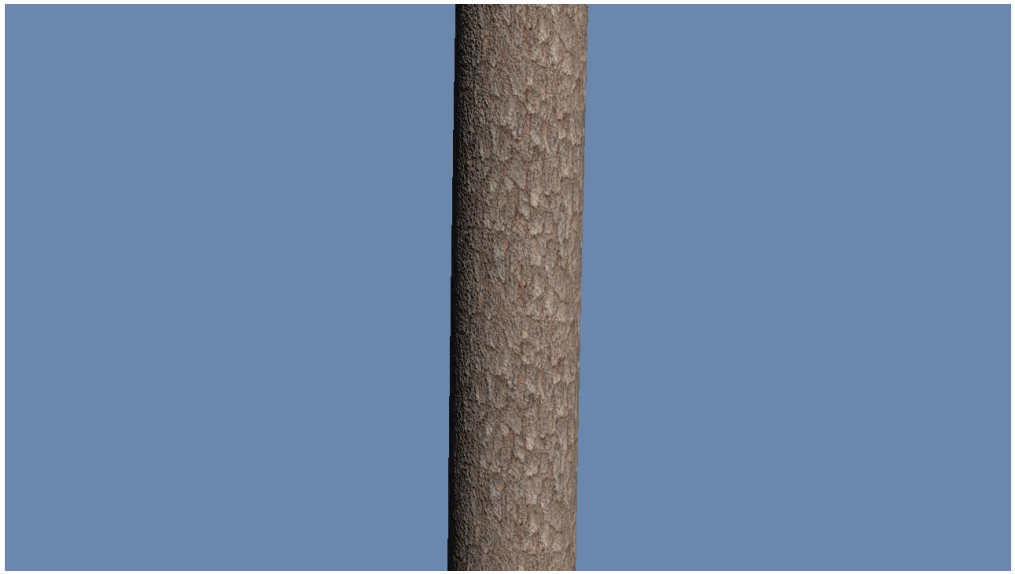


UPDATED



Peeling bark: fibrous strips and woody grain replace the old mottled scanned surface.





Scaly bark: layered gray-brown pine plates replace the orange-speckled scanned surface.

Validation & rendering budget

The changes run in authoring/model generation; the foliage shader and its texture-fetch count are unchanged.



Smooth · generated



Fissured · generated



Scaly · generated



Fibrous · generated



Peeling · generated

11 forms × 2 LODs

Wrap normal/tangent checks pass

Same triangle counts

Compared with the staged tree build

3.33 MiB bark bank

Same five512px BC7 material pairs

Native checks pass

Packaged tree and forest shaders

Published bark pixels passed **330,241** boundary, unit-normal and linear-mip checks. Native scratch-buffer checks ensure generated meshes own their data after cached buffers are reused.

Performance scope: no new draw passes, texture layers or per-frame processing. A live FPS comparison was not recorded while the existing WorldGen session was running; unchanged budgets are not a global FPS guarantee.

Limits: all bark relief is inferred from generated colour, not measured geometry. Branch junction shapes remain the existing procedural architecture. Foliage textures are unchanged; this revision replaces the bark sources only.

Unity — normals at UV seams

Adobe — light cancellation, tiling and normal inputs

Moisan2011 — periodic-plus-smooth image decomposition

Poly Haven — scan reuse terms and attribution

All five master paths and full prompts: assets/verification/tree-bark-generation-prompts-20261107.json. Built-in image_gen; canonical assets backed up in the verified GitHub asset release.